

K way polyphase merge

Monday, December 16, 2024 7:35 PM

$k=3$

T_1 0 1 0 2 6 13 0 24

T_2 0 1 2 0 4 11 24 0

T_3 0 1 2 4 0 7 20 44

T_4 1 0 1 3 7 0 13 37

← merge {runs} into

pass : 1 ← 2 ← 3 ← 4 ← 5 ← 6 ← 7 ← 8

runs on tape : $F_N^3, F_{N-1}^3 + F_{N-2}^3, F_{N-1}^3, 0$

runs before : $0, F_{N+1}^3, F_{N-1}^3 + F_N^3, F_N^3$

runs in total : $F_{N+1}^3 + F_{N-1}^3 = F_{N+2}^3 - F_N^3$

for k-way polyphase merge :

runs on tape :

T_1 F_N^k 0
 T_2 $F_{N+1}^k - F_N^k$ $F_{N+1}^k = F_N^k + F_{N-1}^k + \dots + F_{N-k+1}^k$
 $\vdots$
 T_i $F_{N-1}^k + \dots + F_{N-k+i}^k$ $F_N^k + F_{N-1}^k + \dots + F_{N-k+1}^k + F_{N-k+i}^k$
 $\vdots$
 T_{k+1} 0 F_N^k

runs in total : $\sum_{i=1}^k i F_{N-k+i}^k$